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Yang–Mills, Tracks 1 and 2: area law at $\beta_W < 2/3$ (verified) and a direct Dobrushin mass-gap threshold $\beta_W < 2/9$ (open)

Progress document with consistency and reproducibility record

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adversarial re-derivation and drafting assisted by Claude (Anthropic), 28 Sept. 2026

Tier key (SI standard): (P) proved here or in the cited manuscript; (C) certified by independent computation (this document, script `verify_track2.py`, mpmath 50-digit / exact rationals); (L) literature item still to be located and checked verbatim; (Open) not yet established; (Withdrawn) retracted claim.

1 Scope and status

1. **Track 1** (manuscript "area law SU(2), $\beta_W < 2/3$", v1.1). Proves an *area law* for 4D lattice SU(2) Wilson action at $\beta_W < 2/3$ via the Durhuus–Fröhlich (DF80) slab reduction plus a chordal-W_1 Dobrushin criterion on von Mises–Fisher (vMF) single-site laws. Its numerical certificates (Lemmas 6.1–6.2) are independently re-verified in §3. (P)+(C), with the (L) items of its own §10 still open.
2. Track 1 does *not* yield a mass gap: the DF80 reduction is a one-sided upper bound on $\langle W(C)\rangle$, and SU(2) has no FKG/Griffiths structure to make it two-sided. (P) (structural).
3. **Track 2** (this document, §4). Dobrushin uniqueness applied *directly* to the 4D link variables, with no gauge fixing, using Track 1's Corollary 6.3 as the only analytic input. Candidate result: uniqueness of the DLR state and exponential clustering (hence a fixed-lattice mass gap) for $\beta_W < 2/9$. Status: (Open) pending one (L) item (§4.3); all other steps are (P) below.
4. **What neither track is.** Both are statements at fixed lattice spacing, deep in strong coupling. The SU(2) crossover is $\beta_W \approx 2.2$; the continuum problem requires $\beta_W \to \infty$. Nothing here bears on the Clay statement, and it must not be framed as if it did. A fixed-β mass gap is already a theorem (Osterwalder–Seiler 1978, cluster expansion; Shen–Zhu–Zhu 2023, Bakry–Émery). Track 2's content is a *threshold improvement*: $1/12 \to 2/9$ in Wilson units (§5).

2 Conventions (frozen)

$G = SU(2) \cong S^3 \subset \mathbb{H}$, $U \leftrightarrow q$ unit quaternion; $\frac{1}{2}\Re\,\mathrm{Tr}(U_1U_2U_3U_4) = \langle q_1, \overline{q_2q_3q_4}\rangle$, Euclidean inner product on $\mathbb{R}^4$. Chordal metric $d(q,q') = |q-q'|$, diam $= 2$; a map $q \mapsto \langle q, w\rangle$, $|w| = 1$, is 1-Lipschitz. Wilson action on $\Lambda \subset \mathbb{Z}^4$ (periodic or fixed boundary):

$$\mu_\Lambda(dU) \;\propto\; \exp\Big(\beta_W \sum_{p\in\mathcal{P}_\Lambda^+} \tfrac{1}{2}\Re\,\mathrm{Tr}\,U_p\Big) \prod_{\ell\in E_\Lambda^+} \sigma(dU_\ell), \qquad \beta_W = 4/g^2.$$

vMF law on S^3: $\mu_\eta \propto e^{\eta\cdot n}d\sigma$, $\eta \in \mathbb{R}^4$. Track 1 objects: $L(\eta,v) = \sup\{\mathrm{Cov}_{\mu_\eta}(f, v\cdot n) : f \text{ 1-Lipschitz}\}$; Corollary 6.3: $L(\eta,v) \le \frac{1}{2}\sqrt{\max(A,B)} < \frac{1}{4}$ for $0 < |\eta| \le 4$, $= \frac{1}{4}$ at $\eta = 0$. Conversion to Shen–Zhu–

Zhu (SZZ23) conventions: their action is $S(Q) = N\beta\,\Re\sum_p \operatorname{Tr} Q_p$ (their eq. (1.2)), so $N\beta = \beta_W/N$, i.e. $\beta_W = N^2\beta = 4\beta$ for SU(2). (P)

3 Track 1: independent verification of the analytic certificates

The two Bessel inequalities are, by Prop. 5.2 of the manuscript,

$$F_B(r) = 24r^2 I_1(r)^2\big(\tfrac14 - B_{\text{bound}}(r)\big), \qquad D_A(r) = 12 f(r)^4\big(\tfrac14 - A_{\text{bound}}(r)\big), \quad f = 2I_1(r)/r,$$

which confirms algebraically that the partially obscured last term of Lemma 6.1 is $-r^2 I_2(2r)$. (P)

Item	Manuscript claim	Independent result (mpmath 50 d.p. / sympy exact)	Tier
F_B series	$c_0 = \cdots = c_3 = 0$, $c_4 = 1/1152$, $c_5 = 7/92160$, $c_6 = 1/368640$, $c_7 = 17/371589120$, $c_8 = 1/9909043200$	identical: $0,0,0,0,\frac{1}{1152},\frac{7}{92160},\frac{1}{368640},\frac{17}{371589120},\frac{1}{9909043200}$	C
D_A series	$\frac38 r^2 + \frac{9}{64}r^4 + \frac{143}{5760}r^6 + \ldots$	identical; next terms $\frac{101}{36864}r^8 + \frac{1097}{5160960}r^{10}$ (new)	C
$F_B > 0$ on $(0,4]$	margin 8.47×10^{-4}	$\min_{\text{grid}} F_B/r^8 = 8.680574544287787\ldots\times10^{-4}$ at $r = 0.005$, increasing	C
$D_A > 0$ on $(0,4]$	$D_A \ge r^2(3/8 - 0.0047)$	$\min_{\text{grid}} D_A/r^2 = 0.3750035156405165\ldots$ at $r = 0.005$	C
Cor. 6.3	$L < 1/4$ for $\eta \neq 0$	$\frac12\sqrt{\max(A,B)}$ at $r = 0.3, 1, 2.5, 4$: 0.2499994225, 0.2499382389, 0.2488248890, 0.2473426207	C

Table 1: Grid: $r = k/200$, $k = 1,\ldots,800$. Not reproduced here: the $k > 80$ and $k > 189$ tail-majorant arguments of Lemmas 6.1–6.2 (they remain at the manuscript's own (C)).

Consistency with Fig. "Bound $\frac12\sqrt{c^2A + s^2B}$ vs angle": at $r = 4$, $\frac12\sqrt{A} = 0.1528$, $\frac12\sqrt{B} = 0.2473$, matching the $r = 4$ curve's $0°$ and $90°$ extents.

4 Track 2: direct Dobrushin uniqueness on the 4D link variables

4.1 No gauge fixing is needed

The Wilson measure is a finite-range Gibbs specification on the site set E^+ (links) with compact single-site space S^3. Dobrushin's theorem applies to any such specification; gauge invariance is a symmetry of the specification and only forces the unique state (when it exists) to be gauge-invariant. This is exactly the route used by Chatterjee (CMP 2020) for $\mathbb{Z}_2$ gauge theory. The earlier concern (this thread, first turn) that gauge redundancy obstructs a single-site specification is (Withdrawn).

Lemma 4.1 (single-link conditional law)**.** *Fix a link ℓ and all other links. The conditional law of q_ℓ is μ_η with*

$$\eta = \beta_W \sum_{s\in \mathrm{St}(\ell)} \overline{q_s}, \qquad q_s = \textit{product of the three other link variables of the plaquette (staple)},$$

the sum running over the $2(d-1) = 6$ staples of ℓ. Hence $|\eta| \le 6\beta_W$. (P)

Proof. Every plaquette containing ℓ contributes $\beta_W\,\frac12\Re\operatorname{Tr}(U_\ell U_s) = \beta_W\langle q_\ell, \overline{q_s}\rangle$ by §2; sum over the six plaquettes. □

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Lemma 4.2 (interaction matrix)**.** *Let* $C_{\ell\ell'} = \sup\{W_1(\mu_\ell(\cdot|\omega), \mu_\ell(\cdot|\omega'))/d(\omega_{\ell'}, \omega'_{\ell'})\}$ *over pairs agreeing off* ℓ'*. Then* $C_{\ell\ell'} = 0$ *unless* ℓ, ℓ' *share a plaquette (18 such* ℓ' *in* $d = 4$*), and for those, whenever* $6\beta_W \le 4$*,*

$$C_{\ell\ell'} \le \beta_W \sup_{|\eta|\le 6\beta_W,\ |v|=1} L(\eta, v) \le \beta_W/4.$$

Consequently $\alpha := \sup_\ell \sum_{\ell'} C_{\ell\ell'} \le 18\beta_W/4 = 9\beta_W/2$*.* (P)*, given Track 1 Cor. 6.3.*

Proof. Changing one factor $u \to u'$ of a staple product changes the product by $|u - u'|$ (left/right multiplication by unit quaternions is an isometry of S^3), so $|\delta\eta| = \beta_W|u - u'|$. Along $\eta_t = \eta_0 + t\delta\eta$, Kantorovich duality gives $W_1(\mu_{\eta_0}, \mu_{\eta_1}) \le |\delta\eta| \sup_t L(\eta_t, \hat{\delta})$ (Track 1, §7). Apply Cor. 6.3, valid since $|\eta_t| \le 6\beta_W \le 4$. □

Remark 4.3. Replacing $1/4$ by the true supremum of $\frac{1}{2}\sqrt{\max(A, B)}$ over $|\eta| \le 6\beta_W$ gains essentially nothing: the transverse branch has $B_{\text{bound}}(4/3) = 0.2496529505\ldots$, so $\frac{1}{2}\sqrt{B} = 0.2498264150\ldots$ (C). All slack is in the neighbour count, not in L.

4.2 From the Dobrushin condition to covariance decay

The following is the one step not already in Track 1. It is stated and proved in finite volume; the bound is uniform in Λ and boundary condition, so it passes to every infinite-volume limit point.

Lemma 4.4 (W_1-Dobrushin covariance estimate)**.** *Let* S *be a finite site set,* X_i *compact metric spaces with* $\operatorname{diam} X_i \le D$*,* μ *a probability measure on* $\prod X_i$ *with conditional laws* $\mu_i(\cdot|\omega)$*, and* $C = (C_{ij})$ *the* W_1*-interaction matrix as in Lemma 4.2. Assume* $\alpha = \max_i \sum_j C_{ij} < 1$*. For* f, g *with coordinate Lipschitz constants* $\delta_i(f) = \sup\{|f(\omega) - f(\omega')|/d(\omega_i, \omega'_i) : \omega, \omega' \text{ agree off } i\}$*,*

$$|\operatorname{Cov}_\mu(f, g)| \ \le\ D^2 \sum_{i,k\in S} \delta_k(f)\,\big[(1 - C)^{-1}\big]_{ki}\,\delta_i(g), \qquad \big[(1 - C)^{-1}\big]_{ki} \le \frac{\alpha^{\operatorname{dist}_C(k,i)}}{1 - \alpha},$$

where dist_C *is graph distance in the graph with edges* $\{C_{ij} \ne 0\}$*.* (P)

Proof. Dynamics. Let $Lh = \sum_i(\mu_i h - h)$ be the heat-bath (Glauber) generator, $P_t = e^{tL}$; μ is reversible for P_t because each μ_i is a μ-conditional expectation.

Coupling. Run two copies from ω, ω' with the same Poisson clocks; at a ring of site i resample both from an optimal W_1-coupling of $\mu_i(\cdot|\omega_t)$ and $\mu_i(\cdot|\omega'_t)$ (measurable selection of optimal couplings on compact spaces: Villani, Cor. 5.22 — (L) for the exact statement). By the triangle inequality for W_1 along a chain of single-site changes, $W_1(\mu_i(\cdot|\omega), \mu_i(\cdot|\omega')) \le \sum_j C_{ij} d(\omega_j, \omega'_j)$. Hence $\varphi_i(t) = \mathbb{E}\, d(\omega^i_t, \omega'^i_t)$ satisfies $\dot\varphi \le -(1 - C)\varphi$ componentwise; since $C \ge 0$ entrywise, comparison for cooperative linear systems gives

$$\varphi(t) \le e^{-t(1-C)}\varphi(0), \qquad e^{-t(1-C)} = e^{-t}\sum_{n\ge 0} \frac{t^n C^n}{n!} \ \ge 0.$$

Coordinate Lipschitz constants propagate. For ω, ω' differing only at j: $|P_t f(\omega) - P_t f(\omega')| \le \sum_k \delta_k(f)\varphi_k(t) \le \sum_k \delta_k(f)[e^{-t(1-C)}]_{kj} d(\omega_j, \omega'_j)$, so $\delta_j(P_t f) \le \sum_k \delta_k(f)[e^{-t(1-C)}]_{kj}$. Row sums of $e^{-t(1-C)}$ are $\le e^{-(1-\alpha)t}$, so $P_t f \to \mu(f)$ uniformly and $\int_0^\infty LP_t f\, dt = \mu(f) - f$.

Covariance. $\operatorname{Cov}(f, g) = \mathbb{E}_\mu[g(f - \mu f)] = -\int_0^\infty \mathbb{E}_\mu[g\, LP_t f]\, dt = \int_0^\infty \sum_i \mathbb{E}_\mu\big[(g - \mu_i g)(P_t f - \mu_i P_t f)\big] dt$, the last step because $\mathbb{E}_\mu[\mu_i g \cdot (h - \mu_i h)] = \mathbb{E}_\mu[\mu_i g \cdot \mu_i(h - \mu_i h)] = 0$. Pointwise $|h - \mu_i h| \le$

$\delta_i(h) \int d(\omega_i, x)\mu_i(dx|\omega) \le D\,\delta_i(h)$. Therefore

$$|\operatorname{Cov}(f,g)| \le D^2 \sum_i \delta_i(g) \int_0^\infty \delta_i(P_t f)\,dt \le D^2 \sum_{i,k} \delta_k(f)\,\delta_i(g) \int_0^\infty [e^{-t(1-C)}]_{ki} dt$$
$$= D^2 \sum_{i,k} \delta_k(f)[(1-C)^{-1}]_{ki}\delta_i(g).$$

Finally $(1-C)^{-1} = \sum_n C^n$ entrywise, $[C^n]_{ki} = 0$ for $n < \operatorname{dist}_C(k,i)$, and $[C^n]_{ki} \le \sum_i [C^n]_{ki} \le \alpha^n$. □

Lemma 4.5 (graph distance vs. lattice distance)**.** *For links ℓ, ℓ' sharing a plaquette, the midpoints satisfy $\|m_\ell - m_{\ell'}\|_\infty \le 1$. Hence $\operatorname{dist}_C(\ell,\ell') \ge \|m_\ell - m_{\ell'}\|_\infty \ge \frac{1}{2}\|m_\ell - m_{\ell'}\|_2$ in $d = 4$.* **(P)**

Proof. In the (μ,ν) plaquette at x the midpoint differences are $\pm\frac{1}{2}e_\mu \pm \frac{1}{2}e_\nu$ (adjacent sides) or $\pm e_\nu$ (opposite sides). □

4.3 Candidate theorem

Theorem 4.6 (Track 2, candidate)**.** *Let $d = 4$, $G = SU(2)$, Wilson action, $\beta_W < 2/9$, $\alpha = 9\beta_W/2 < 1$. Then, uniformly in Λ and boundary condition:*

(i) For plaquette observables $f_p = \frac{1}{2}\Re \operatorname{Tr} U_p$ ($\delta_\ell(f_p) = 1$ on the four links of p, 0 otherwise),

$$|\operatorname{Cov}_{\mu_\Lambda}(f_p, f_{p'})| \;\le\; \frac{64\,\alpha^{\|p-p'\|_\infty - 1}}{1-\alpha},$$

where $\|p-p'\|_\infty$ is the ℓ^∞ distance between plaquette centres. More generally $|\operatorname{Cov}(f,g)| \le 4\sum \delta_k(f)\alpha^{\operatorname{dist}_C(k,i)}\delta_i(g)/(1-\alpha)$ for all Lipschitz cylinder f, g.

(ii) The infinite-volume Gibbs state is unique and translation invariant (Dobrushin uniqueness, W_1 form). **(L)***: cite Dobrushin 1970 / Dobrushin–Shlosman 1985 / Wu 2006 for the metric-space version.*

(iii) Given (i), (ii) and Osterwalder–Seiler reflection positivity, the transfer matrix T on the physical Hilbert space has $\operatorname{spec}(T) \cap (\alpha, 1) = \emptyset$, i.e. mass gap $m \ge -\log(9\beta_W/2)$ in lattice units. **(P)** *modulo (ii) and the standard density argument (Lipschitz cylinder functions are dense; spectral mass in $(\alpha+\varepsilon, 1)$ would contradict the uniform rate).*

Proof. (i) is Lemma 4.4 with $D = 2$, C from Lemma 4.2, and Lemma 4.5 (plaquette-centre ℓ^∞ separation T forces link-midpoint separation $\ge T - 1$); $4 \cdot 16 = 64$. (ii), (iii) as indicated. □

Overall tier for Theorem 4.6: (Open), to become (P) once the two (L) items (measurable optimal coupling; metric-space Dobrushin uniqueness) are matched to verbatim statements. No further calculation is required.

5 Comparison, sharpness, sanity

Sanity check against the leading strong-coupling glueball estimate $m\,a \approx -4\ln u$, $u = I_2(\beta_W)/I_1(\beta_W)$ (heuristic, non-rigorous; comparison only). (C)

β_W	0.05	0.10	0.15	0.20	0.22
$-\log(9\beta_W/2)$	1.4916548768	0.7985076962	0.3930425881	0.1053605157	0.0100503359
$-4\ln(I_2/I_1)$	17.5285231620	14.7571837890	13.1374038725	11.9895846726	11.6097388142

Result	Method	Threshold (Wilson β_W, $d = 4$, SU(2))	Tier
Osterwalder–Seiler 1978	cluster expansion	small, not explicit	L
Shen–Zhu–Zhu, CMP 400 (2023)	Bakry–Émery / Langevin coupling	$\beta < 1/(16(d-1)) = 1/48 \Rightarrow \beta_W < 1/12 = 0.0833\ldots$	L (abstract + eq. (1.2) inspected)
Track 2 (this document)	single-site chordal-W_1 Dobrushin	$\beta_W < 2/9 = 0.2222\ldots$ (ratio 8/3)	Open
Track 1 area law	DF80 slab + same Dobrushin	$\beta_W < 2/3$ (area law only)	P

The rigorous lower bound lies far below the expected gap and never crosses it; it vanishes at threshold, as any Dobrushin-type bound must. Consistent; no contradiction.

Why 2/9 and why not more. Track 1's Theorem C shows the transverse vMF branch has $L \to 1/4$ as $\eta \to 0$, so any single-site chordal-W_1 criterion is limited to $\alpha \leq (\text{neighbours}) \cdot \beta_W/4$: $6\beta_W/4$ in the 3D slab (2/3), $18\beta_W/4$ on 4D links (2/9). Passing 2/9 requires either a direction-aware (transverse/longitudinal) coupling or a Dobrushin–Shlosman block criterion (e.g. cube or plaquette-star blocks). Both are (Open).

6 Failure modes (hostile-review pass)

1. *Circularity.* Lemma 4.2 uses Cor. 6.3 at $|\eta| \leq 4/3$ only; Track 1 certifies it on $(0, 4]$. No circularity; the certificates are over-sufficient here.
2. *W_1 vs TV.* The classical Dobrushin/Georgii statements are in total variation; Lemma 4.4 is proved here from scratch for the W_1 matrix, so the argument does not depend on the (L) item for (i). It does for (ii) (uniqueness), where the W_1 version is standard but must be cited exactly.
3. *Infinite volume.* All bounds are uniform in Λ; the dynamics argument is finite-volume only, deliberately, to avoid constructing infinite-volume Glauber dynamics.
4. *Mass gap definition.* (iii) is the transfer-matrix gap in the OS Hilbert space; if the intended notion is the decay rate of *all* connected correlations of gauge-invariant observables, (i) already gives it with rate $-\log \alpha$ per unit ℓ^∞ distance.
5. *Constant* 64. Crude (diam$^2 \times 16$ link pairs); irrelevant to the rate.
6. *Strong-coupling comparison* is heuristic and used only to check the sign/order of the lower bound.
7. *Not the continuum problem* (§1, item 4).

7 Verification record

Adversarial re-derivation, 28 Sept. 2026, script `verify_track2.py` (mpmath 50 d.p.; sympy exact rationals; IEEE-754 doubles never used for certified quantities):

C1 Lemmas 6.1–6.2 grid positivity and reconstructed A, B bounds: as in §3.

C2 Exact series coefficients of F_B (to r^{16}) and D_A (to r^{10}): all manuscript values reproduced; two new D_A coefficients recorded.

T2 Neighbour count 18, threshold 2/9, SZZ conversion 1/12, ratio 8/3, gap table above.

- Sources inspected verbatim: SZZ23 abstract and eq. (1.2) (ar5iv 2204.12737); Chatterjee CMP 2020 §5.1 (Dobrushin ⇒ decay, TV version, with his note that no exact literature statement was found). Not inspected: OS78 threshold; DF80 p. 147; Villani Cor. 5.22 wording; Wu 2006 statement.
- Corrections this pass: withdrawal of the "gauge fixing needed" concern; scope statement (§1, item 4) made explicit.

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8 Checklist

Item	Type	Action
Villani Cor. 5.22 (measurable optimal coupling)	L	quote statement; confirm applies to S^3, chordal metric
W_1-Dobrushin uniqueness (Dobrushin 70; D–S 85; Wu 06)	L	locate metric-space version used for (ii)
OS78 transfer matrix / RP for Wilson action	L	confirm Hilbert space and density of Lipschitz cylinders
Lemma 4.4	P	second reader; check $\mathbb{E}_\mu[\mu_i g(h-\mu_i h)]=0$ step and cooperative comparison
Lemma 4.5	P	trivial; keep
Block (Dobrushin–Shlosman) extension past 2/9	Open	decide block shape; compute C for plaquette-star
Direction-aware coupling	Open	exploit $A<B$ (longitudinal $\ll$ transverse) in the coupling

A Script `verify_track2.py`

```
#!/usr/bin/env python3
# verify_track2.py  --  independent re-derivation for SI Yang-Mills Tracks 1/2 (28 Sept 2026)
# Requires: mpmath, sympy.  All arithmetic mpmath at 50 digits or exact rationals; no rounding.
from mpmath import mp, mpf, besseli, diff, log, sqrt
import sympy as sp
mp.dps = 50
I = lambda n, x: besseli(n, x)

# ---- Lemma 6.1 / 6.2 reconstruction from Prop 5.2 -------------------------
def FB(r):  # = 24 r^2 I1^2 * (1/4 - B_bound)
    return 6*r**2*I(1, r)**2 - 64*I(2, r)**2 - 96*I(3, r)**2 - r**2*I(2, 2*r)
f = lambda r: 2*I(1, r)/r
def DA(r):  # = 12 f^4 * (1/4 - A_bound)
    f0, f1, f2, f3 = f(r), diff(f, r), diff(f, r, 2), diff(f, r, 3)
    return (3*f0**4 - 32*(f2*f0 - f1**2)**2 - 32*(f3*f0 - f2*f1)**2
            - 4*(diff(f, 2*r, 2)*f0**2 - 2*f0*f1*diff(f, 2*r) + f1**2*f(2*r)))

rs = [mpf(k)/200 for k in range(1, 801)]                # grid on (0,4], step 0.005
fb = [FB(r)/r**8 for r in rs]; da = [DA(r)/r**2 for r in rs]
print("[C1] min F_B/r^8 on grid:", min(fb), "at r =", rs[fb.index(min(fb))])
print("[C1] min D_A/r^2 on grid:", min(da), "at r =", rs[da.index(min(da))])
for r in [mpf('0.3'), mpf(1), mpf('2.5'), mpf(4), mpf(4)/3]:
    Bb = mpf(1)/4 - FB(r)/(24*r**2*I(1, r)**2)
    Ab = mpf(1)/4 - DA(r)/(12*f(r)**4)
    print(f"[C1] r={r}: B_bound={Bb}  A_bound={Ab}  L_bound=1/2 sqrt(max)={sqrt(max(Ab,Bb))/2}")

# exact series coefficients (sympy)
r = sp.symbols('r')
sFB = sp.series(6*r**2*sp.besseli(1, r)**2 - 64*sp.besseli(2, r)**2 - 96*sp.besseli(3, r)**2
                - r**2*sp.besseli(2, 2*r), r, 0, 18).removeO()
print("[C2] F_B coefficients:", [(k, sFB.coeff(r, k)) for k in range(0, 18, 2)])
N = 12
fs = sum((r/2)**(2*k)/(sp.factorial(k)*sp.factorial(k+1)) for k in range(N))
f1, f2, f3 = fs.diff(r), fs.diff(r, 2), fs.diff(r, 3)
DAs = (3*fs**4 - 32*(f2*fs - f1**2)**2 - 32*(f3*fs - f2*f1)**2
       - 4*(f2.subs(r, 2*r)*fs**2 - 2*fs*f1*f1.subs(r, 2*r) + f1**2*fs.subs(r, 2*r)))
p = sp.Poly(sp.expand(DAs), r)
print("[C2] D_A coefficients:", [(k, p.coeff_monomial(r**k)) for k in range(0, 12, 2)])

# ---- Track 2: direct 4D Dobrushin -----------------------------------------
d = 4; nb = 2*(d-1)*3
beta_star = mpf(4)/nb
print("[T2] neighbours per link =", nb, "; beta_* = 4/nb =", beta_star)
print("[T2] SZZ23 SU(2) threshold, Wilson units 4/48 =", mpf(4)/48, "; ratio =", beta_star/(mpf(4)/48))
print("[T2] beta_W  alpha=9beta_W/2  -log(alpha)  -4 ln(I2/I1) (strong-coupling glueball, heuristic)")
for b in [mpf('0.05'), mpf('0.1'), mpf('0.15'), mpf('0.2'), mpf('0.22')]:
```

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```
a = 9*b/2; u = I(2, b)/I(1, b)
print("[T2]", b, a, -log(a), -4*log(u))
```